

Improving Downstream Task Performance in Multilingual LLMs by Intervening Language Specific Neurons

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Introduction

Problem Statement

● Objective:

- Investigate the representation of language-specific information in large language models (LLMs).
- Develop methods to identify and utilize language-specific neurons for multilingual tasks.
- Enhance performance for low-resource languages by employing techniques such as activation probability entropy and LoRA-based fine-tuning.

● Importance:

- Ensure fair and inclusive NLP systems by bridging the performance gap between high-resource and low-resource languages.
- Advance understanding of the internal mechanisms of multilingual LLMs.
- Contribute to the development of efficient methods for improving multilingual performance.

Presentation Outline

- Review of existing methods for identifying and utilizing language-specific neurons.
- Experimental design: fine-tuning configurations and intervention strategies.
- Results and analysis for multilingual tasks (e.g., XNLI task).
- Conclusion and future work directions.

Methodology: Definition of Neuron

What is a Neuron?

- A **neuron** corresponds to a specific dimension of the activation vector $\mathbf{h}^i$, represented as:

$$\mathbf{h}^i = [\text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \otimes (\hat{\mathbf{H}}^i \mathbf{W}_3^i)] \cdot \mathbf{W}_2^i, \quad (1)$$

- The average activation (against Wikipedia sentence) of a neuron (i, j) across token positions is computed as:

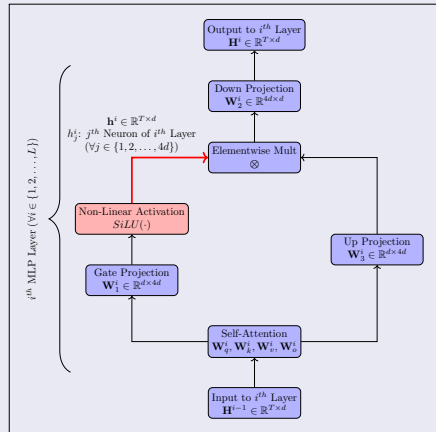
$$\bar{h}_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t. \quad (2)$$

- Language-specific activations are computed as:

$$h_{i,j}^l = \frac{1}{|S_l|} \sum_{s \in S_l} \bar{h}_{i,j}^{(s)}, \quad (3)$$

where S_l is the set of all sequences for language l .

MLP Architecture of Llama Model

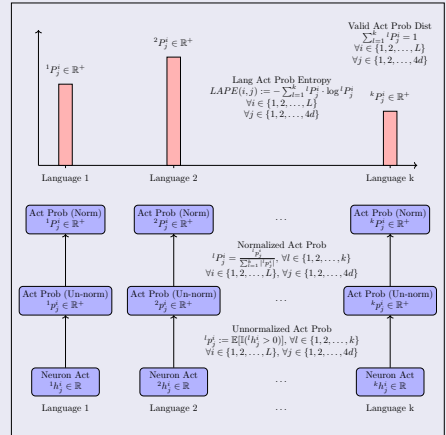


Methodology: LAPE

What is a LAPE?

• A

LAPE Workflow



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Thank You!